

ECON696 - Independent Study

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Motivation

As artificial intelligence (AI) tools rapidly expand across industries, concern is growing about their impact on the labor market, especially for those starting their careers. A common fear is that AI may be automating tasks traditionally performed by entry-level workers, effectively narrowing the bottom rung of the career ladder.

Entry-level positions are often the first point of access into an occupation or industry, particularly for recent graduates and those switching careers. If these positions are declining disproportionately in AI-exposed fields, it could signal growing barriers to entry for large segments of the workforce. This study investigates that possibility by measuring whether the share of entry-level job openings has declined more sharply in AI-exposed occupations than in others over the past decade.

Data

This study uses Lightcast job postings data from 2014 to 2024. Each job posting contains rich data, including:

- Occupation code
- Posting date
- Minimum required education and experience
- Skills listed in the job description

The dataset represents real-time job postings across industries in the US, capturing employer demand trends over time. It covers approximately 92% of all job postings in the US. To ensure consistency, only occupations that appeared every year from 2014 to 2024 and had a sufficient number of postings were included.

Methodology

The main idea of this independent study is to track the share of postings in each occupation that qualify as entry-level over time, and then compare the trends between occupations that are highly exposed to AI and those that are not.

The analysis is structured around four main steps:

1. Define entry-level job postings:

A job posting is considered *entry-level* if it meets the following criteria:

- Requires 1 year or less of experience OR seniority level is junior and year of experience has a null value
 - Does not require graduate education (requires at most a bachelor's degree or equivalent; specifically: "No Education Listed", "High school or GED", "Associate degree", or "Bachelor's Degree")
2. Compute entry-level share per occupation-year

For each occupation (based on 5-digit SOC 2021 codes) and each year from 2014 to 2024, the entry-level share is calculated as:

$$\text{Entry-Level Share} = \frac{\text{Entry-Level Postings}}{\text{Total Postings}}$$

3. Define AI Exposure per Occupation:

AI exposure is computed using the share of job postings from 2014 to 2024 that mention key AI-related skills. If a significant portion of postings within an occupation include these skills, it is inferred that the occupation is likely to be impacted by AI tools and workflows.

The following AI-related keywords were used to flag postings as AI-relevant:

Machine Learning, Artificial Intelligence, Computer Science, Data Science, Software Engineering, Software Development, Algorithms, Generative Artificial Intelligence, Apache Spark, Java (Programming Language), Big Data, Data Engineering, C++ (Programming Language), Data Management, Predictive Modeling, Extract Transform Load (ETL), DevOps, Deep Learning, JavaScript (Programming Language), Data Pipelines, Kubernetes, Data Warehousing

An AI exposure percentage was calculated for each occupation as:

$$\text{AI exposure rate} = \frac{\text{AI related postings}}{\text{Total Postings in}}$$

4. Classify Occupations as AI-Exposed or Not:

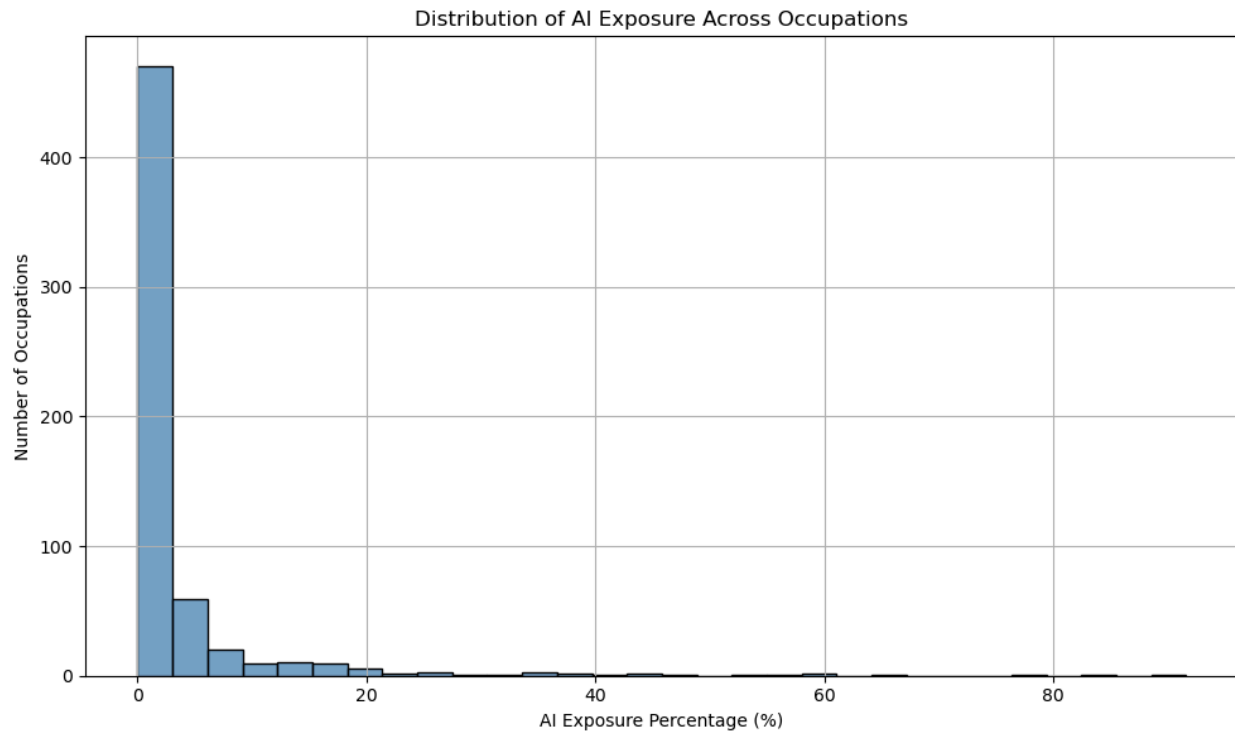
A threshold of 4.8% AI exposure was selected as the cutoff. Occupations with a AI exposure percentage equal to or above 4.8% were labeled AI-exposed, the rest were labeled non-AI-exposed.

5. Compute Weighted Entry-Level Shares:

For each year, the overall entry-level share in both AI and non-AI groups was calculated

using a weighted average, where each occupation's entry-level share is weighted by its number of postings.

Findings



Above is a histogram of AI exposure rate on x axis and number of occupations on the y axis. The median AI exposure is 0.65%, but a threshold of 4.8% was set to define AI-exposed occupations in order to focus the analysis on roles with a meaningful concentration of AI-related skills in job postings, rather than capturing low-level exposure.

Below are the top 20 occupations ranked by AI exposure rate:

The occupations with the highest AI exposure rates are overwhelmingly concentrated in the tech sector, particularly in roles related to software development, data management, and computer systems. For example, over 90% of job postings for *Computer and Information Research Scientists* mention AI-related skills, followed by high exposure rates for *Web Developers* (83.5%), *Software Developers* (78.1%), and *Database Administrators* (64.4%).

soc_2021_5_name	postings_2014	postings_2024	posting_growth_pct	ai_postings	total_postings	ai_exposure_pct
Computer and Information Research Scientists	8380	54517	550.56	318363	347580	91.59
Web Developers	118700	45319	-61.82	1072094	1284001	83.50
Software Developers	616343	548525	-11.00	7424577	9504113	78.12
Database Administrators	99677	86999	-12.72	848707	1318579	64.37
Software Quality Assurance Analysts and Testers	86473	56385	-34.79	644145	1062233	60.64
Database Architects	83467	81192	-2.73	647138	1095618	59.07
Computer Programmers	47249	20834	-55.91	244902	426840	57.38
Data Scientists	184687	230086	24.58	1542899	2886911	53.44
Computer Occupations, All Other	311785	412911	32.43	2515987	5266042	47.78
Web and Digital Interface Designers	81573	58845	-27.86	498811	1091752	45.69
Statisticians	13238	9525	-28.05	74341	164358	45.23
Computer and Information Systems Managers	22388	25337	13.17	133603	321492	41.56
Computer Network Architects	135387	103659	-23.44	622401	1637875	38.00
Information Security Analysts	24910	29300	17.62	152953	402820	37.97
Network and Computer Systems Administrators	134026	97755	-27.06	546054	1565635	34.88
Computer Systems Analysts	146436	114895	-21.54	571450	1673254	34.15
Physical Scientists, All Other	2220	2299	3.56	8898	26448	33.64
Computer Hardware Engineers	13169	15893	20.68	68122	209627	32.50
Aerospace Engineers	4969	8843	77.96	26870	93470	28.75
Management Analysts	187747	135389	-27.89	610903	2279960	26.79

Lastly, the figure below compares the share of entry-level job postings between AI-exposed occupations and non-AI-exposed occupations over time. While non-AI-exposed occupations have seen a gradual increase in the share of entry-level postings, rising from approximately 12% to over 16%, AI-exposed occupations have remained flat at around 5%.

My results show that entry-level access to AI-exposed occupations has stagnated, in contrast to the increasing share of entry-level opportunities in non-AI fields. That could be due to higher technical skill requirements, fewer junior roles, or automation of entry-level tasks. This pattern could have long-term implications for economic mobility and workforce equity, particularly if access to high growth, high-wage AI careers becomes concentrated among those with advanced education or prior experience.

